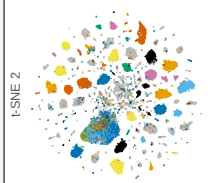
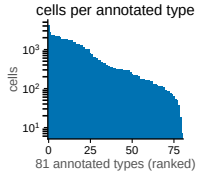
Gene-based baseline: Scanpy clusters + Connectome Model rule + connectome-guided matching

1 Cells



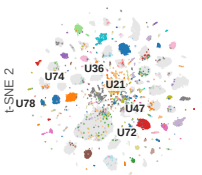
cells per annotated type



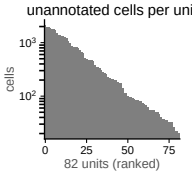
(subsample of 30,000 of 99,656 cells shown)

59,715 annotated cells keep their label (81 types, coloured); 39,941 unannotated cells (grey) have no cell type

2 Clustering



unannotated cells per unit

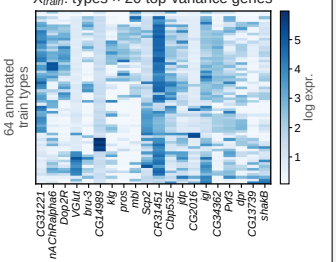


annotated cells light grey; labels: the 6 units matched to most test types (step 4)

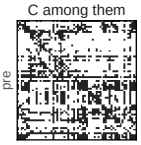
Scanpy Leiden (res 4.0): 106 clusters; 82 with ≥20 unannotated cells become units U0–U81 (cell type unknown)

3 Profiles + gene rule

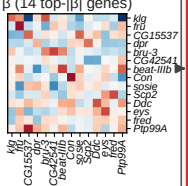
X_{train}: types × 20 top-variance genes

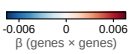


C among them



β (14 top-|β| genes)





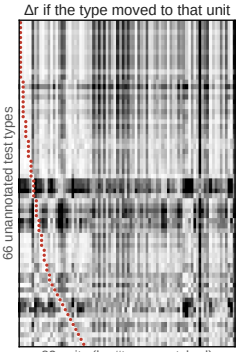
$\hat{C}(a \to X) = x_a\, \beta\, x_X^{\top}$

profile = mean log-expression of the type's cells (2,000 HVGs); ridge β fitted on the 64 ANNOTATED train types only; λ = 642 (= $10^{-6} \cdot s^4_{max}$, chosen on an inner 80/20 split of those types)

4 Matching

uses held-out C

Δr if the type moved to that unit





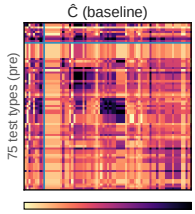
types sharing one unit (max 12):

- U74 ← AOTU065, LC44, LPT04_HST +9
- U47 ← LC46, LT86, LTe35 +5
- U21 ← LTe25, LTe66, VS1 +1
- U36 ← LPi08, LPi10, LT84 +1
- U52 ← Li124_56, MTe13, MTe40 +1
- U72 ← MeTu3c, Sm06, Sm07 +1

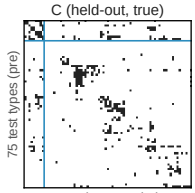
each of the 66 unannotated test types gets ONE unit, chosen to maximize r with the held-out C → 25 distinct units; 9 annotated test types use their own profile

5 Predict + score

$\hat{C}$ (baseline)



C (held-out, true)



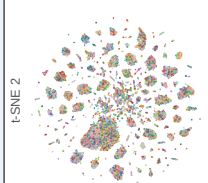
r = 0.396

(no matching: r = 0.029)

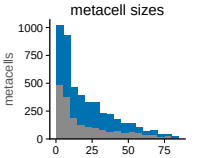
$\hat{C} = x\, \beta\, x^{T}$ for every test pair (rows, columns: the same 75 test types); r($\hat{C}$, C) over all 5,625 entries

ConnectionMiner

1 Metacells



metacell sizes

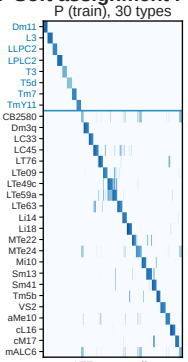


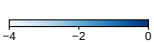
colour = metacell (20 colours recycled); blue bars: mostly-annotated metacells, grey: rest

cells aggregated into 4,748 metacells (train block 4,306, test block 1,960); annotated metacells pinned to their type

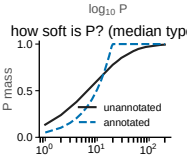
2 Soft assignment P

P (train), 30 types





how soft is P? (median type)

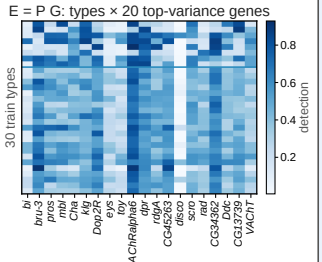


blue names: annotated types; P fixed on their own metacells

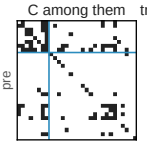
each type's row of P spreads over its candidate metacells (train block: 592 types × 4,306 metacells; subset shown)

3 Profiles + gene rule β

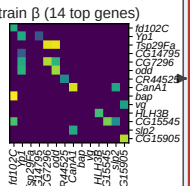
E = P G: types × 20 top-variance genes

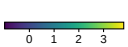


C among them



train β (14 top genes)





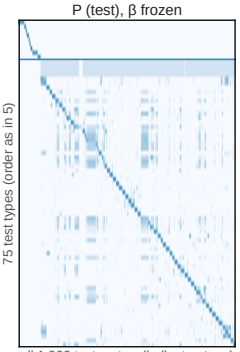
$\hat{C} = E\, \beta\, E^{T} = P\, G\, \beta\, G^{T}\, P^{T}$

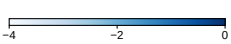
profile E = P G: metacell detection rates weighted by P; β learned jointly with P on ALL 592 train types (64 annotated), 2,997 genes; same 30 types as in 2

4 Test types

uses held-out C

P (test), β frozen



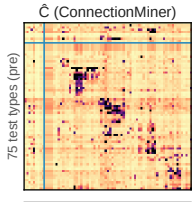


first 9 rows: annotated types, pinned to their own metacells. Effective #metacells (1/Σ P²) of the 66 unannotated test types: median 44 (range 30–1718; baseline: ONE unit per type)

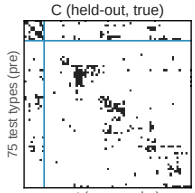
β frozen (train β); P solved for the 75 test types on the held-out C; soft: many metacells per type (median effective #metacells 44)

5 Predict + score

$\hat{C}$ (ConnectionMiner)



C (held-out, true)



r = 0.704

(solver seed 0)

$\hat{C} = P\, G\, \beta\, G^{T}\, P^{T}$ on the test types (same 75 types and order as above); r($\hat{C}$, C) over all 5,625 entries

Visual system, fold 0, test block (75 types: 9 annotated, 66 unannotated); blue lines separate annotated (first) from unannotated types. Red frame: the step uses the held-out connectome.